

Here's a comprehensive Markdown document you can use as your project's README.md or software specification.

Smart TASMAL Consumer Regulation System

«A Digital Platform for Responsible Alcohol Consumption, Consumer Self-Regulation, and Government Analytics»

Project Overview

The Smart TASMAL Consumer Regulation System is a modern web application designed to help regulate alcohol purchases through secure identity verification, personalized consumption limits, and government analytics.

The system allows consumers to create a profile, verify their identity, set drinking limits, enable self-restriction, or declare themselves as teetotalers. Every purchase is securely recorded, analyzed, and monitored to encourage responsible alcohol consumption.

Objectives

- Prevent underage alcohol purchases.
- Promote responsible drinking.
- Allow consumers to control their alcohol consumption.
- Enable government monitoring and analytics.
- Support doctors and de-addiction centres with anonymous statistical reports.
- Generate insights using charts and reports.

Technology Stack

Frontend

- React.js
- TypeScript
- Tailwind CSS
- React Router
- React Query
- Axios
- React Hook Form
- Zod Validation
- Recharts / Chart.js

Backend

- FastAPI
- Python 3.12+
- SQLAlchemy ORM
- Alembic
- Pydantic
- JWT Authentication
- REST API
- CORS Middleware

Database

- PostgreSQL

Authentication

- JWT Token Authentication
- Role-Based Access Control (RBAC)
- Refresh Tokens
- Password Hashing (bcrypt)

Additional Libraries

- QR Code Reader
- QR Code Generator
- PDF Report Generator
- JSON APIs
- Matplotlib
- Pandas
- OpenPyXL (Excel Export)

User Roles

Consumer

- Register
- Login

- Aadhaar Verification (Prototype/Demo)
- Profile Management
- Set Daily Drinking Limit
- Set Weekly Drinking Limit
- Set Monthly Drinking Limit
- Enable Self Restriction
- Enable Teetotaler Mode
- Purchase History
- QR Code Profile
- Download Reports (PDF)

TASMAC Shop Operator

- Login
- Scan Consumer QR Code
- Verify Consumer
- View Purchase Eligibility
- Record Purchase
- Generate Invoice
- Daily Sales Report

Government Administrator

- Dashboard
- Consumer Analytics
- District-wise Statistics
- Revenue Reports

- Shop Management
- User Management
- Alcohol Consumption Reports
- Health Statistics

Doctor Dashboard

- Anonymous Statistics
- Alcohol Consumption Trends
- Risk Analysis
- Health Reports
- Awareness Recommendations

Caretaker Dashboard

- View Consumer Status (with consent)
- Health Notifications
- Restriction Alerts

Core Features

Authentication

- JWT Login
- Refresh Token

- Secure Logout
- Protected Routes

Consumer Profile

- Name
- Age
- Aadhaar ID (Demo)
- Mobile Number
- Address
- Blood Group
- Emergency Contact

Drinking Preference

Users can select:

- Daily Limit
- Weekly Limit
- Monthly Limit

Examples

Beer

- 330 ml
- 660 ml

Wine

- 150 ml

- 300 ml

Spirits

- 30 ml

- 60 ml

Self Restriction

Consumers can

- Stop purchasing for 7 Days

- Stop purchasing for 30 Days

- Stop purchasing for 90 Days

- Permanent Restriction

Teetotaler Mode

Consumer declares

«"I do not consume alcohol."»

System blocks future purchases until the restriction is removed according to policy.

Purchase Monitoring

Each purchase stores

- Consumer ID
- Shop ID
- Product
- Quantity (ml)
- Date
- Time
- Remaining Limit

QR Code System

Generate QR Code for

- Consumer ID
- Purchase Verification
- Digital Membership

Shop Operator

- Scan QR
- Verify User
- Record Purchase

PDF Generator

Generate

- Purchase History
- Monthly Report
- Annual Report
- Government Reports
- Medical Reports
- Revenue Reports

Charts & Analytics

Consumer Dashboard

- Daily Consumption
- Weekly Consumption
- Monthly Consumption

Government Dashboard

- District-wise Sales
- Age Group Analysis
- Revenue Analysis
- Consumption Trends

Doctor Dashboard

- Alcohol Risk Levels
- Health Impact Analysis
- Addiction Statistics

Libraries

- Matplotlib
- Recharts

Reports

Consumer Reports

- Purchase History
- Remaining Limit
- Drinking Statistics

Government Reports

- Revenue
- Shop Performance
- District Analysis
- Age Analysis

Doctor Reports

- Addiction Trends
- Health Impact

- Anonymous Statistics

Exports

- PDF

- Excel

- JSON

API Structure

Authentication

POST /api/auth/register

POST /api/auth/login

POST /api/auth/logout

GET /api/auth/profile

Consumer

GET /api/consumer

PUT /api/consumer/profile

GET /api/consumer/history

POST /api/consumer/restriction

Purchases

POST /api/purchase

GET /api/purchase/history

GET /api/purchase/report

Government

GET /api/admin/dashboard

GET /api/admin/reports

GET /api/admin/analytics

Doctors

GET /api/doctor/statistics

GET /api/doctor/reports

Database Tables

- Users
- Roles
- Consumers
- Shops
- Purchases
- Products
- Restrictions
- Notifications
- Reports
- Doctors
- Caretakers
- QR Codes
- Audit Logs

Responsive Design

Desktop

- Full Dashboard
- Side Navigation
- Advanced Analytics

Tablet

- Responsive Sidebar

- Touch-Friendly UI

Mobile

- Bottom Navigation
- QR Scanner
- Quick Purchase History
- Notifications

Security

- JWT Authentication
- Password Encryption
- HTTPS
- Role-Based Authorization
- SQL Injection Protection
- XSS Protection
- CSRF Protection
- Audit Logs

Future Enhancements

- AI-based Addiction Prediction
- Voice Assistant
- Mobile App (Android/iOS)
- SMS & Email Alerts
- Biometric Authentication

- Integration with Health Services
- Predictive Analytics
- Multilingual Support (Tamil & English)

Expected Outcomes

- Promote responsible alcohol consumption.
- Improve transparency in alcohol sales.
- Support data-driven government policies.
- Provide anonymous insights for healthcare planning.
- Reduce alcohol misuse through voluntary self-regulation.

Project Architecture

React.js (TypeScript + Tailwind CSS)

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FastAPI REST API

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JWT Authentication

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PostgreSQL Database



Matplotlib Reports + PDF Generator + QR Code Module



Government Dashboard | Consumer Dashboard | Doctor Dashboard | Caretaker Dashboard

License

Academic Project

This project is intended for educational and research purposes. Any real-world implementation involving Aadhaar or personal identity data must comply with applicable Indian laws and privacy regulations.